higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

COMPUTER PROGRAMMING
(Second Paper)
NQF LEVEL 4

28 NOVEMBER 2019

This marking guideline consists of 9 pages.

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QUESTION 1

1.1 1.1.1

```
Public Class clsCalculateSession
   Private intDistance1 As Integer
   Private intDistance2 As Integer
   Private intTime1 As Integer
   Private intTime2 As Integer
   Private strBestSession As String
```

(Any 1 of the properties) (1)

1.1.2

```
Sub New(ByVal _time1 As Integer, _
      ByVal _time2 As Integer, _
      ByVal _distance1 As Integer, _
      ByVal _distance2 As Integer)✓
   intTime1 = _time1
   intTime2 = _time2
   intDistance1 = _distance1
   intDistance2 = _distance2✓
End Sub
```

(2)

1.1.3

```
Public Property Distance1() As Integer
   Get
      Return intdistance1
   End Get
   Set(ByVal value As Integer)
      intDistance1 = value
   End Set
End Property

Public Property Distance2() As Integer
   Get
      Return intDistance2
   End Get
   Set(ByVal value As Integer)
      intDistance2 = value
   End Set
End Property

Public Property Time1() As Integer
   Get
      Return intTime1
   End Get
   Set(ByVal value As Integer)
      intTime1 = value
   End Set
End Property
Public Property Time2() As Integer
   Get
      Return intTime2
   End Get
   Set(ByVal value As Integer)
      intTime2 = value
   End Set
End Property

Public Property CalcBestSession() As String
   Get
      Return strBestSession
   End Get
   Set(ByVal value As String)
      strBestSession = value
   End Set
End Property
```

(Any 2 of the properties) (2)

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1.1.4
```
Public Function CalDistance() As Integer✓
    Dim totaldistance As Integer
    totaldistance = intdistance1 + intdistance2✓
    Return totaldistance
End Function
```
(2)

1.1.5
```
Public Function calAverageDistance() As Integer
    Dim Averagedist As Integer
    Averagedist = CalDistance() / 2✓
    Return Averagedist✓
End Function
```
(2)

1.1.6
```
Public Function calTotalTime() As Integer
    Dim totalTime As Integer
    totalTime = inttime1 + inttime2✓
    Return totalTime✓
End Function
End Class
```
(2)

1.1.7
```
Public Function BestSession() As String
    Dim KmPerMin1 As Decimal = 0
    Dim KmPerMin2 As Decimal = 0

    If Time1 <> 0 Then
       KmPerMin1 = Math.Round(Distance1 / Time1, 2)
    End If

    If Time2 <> 0 Then
       KmPerMin2 = Math.Round(Distance2 / Time2, 2)
    End If

    if KmPerMin1 = KmPerMin2 Then ✓
        Return "Week 1 and Week 2 are the same: " &
        KmPerMin1 & " Km/Min"
    elseif KmPerMin1 > KmPerMin1 Then
        Return "Week 1 was the best session: " &
        KmPerMin1 & " Km/Min"✓
    else
        Return "Week 2 was the best session: " &
        KmPerMin2 & " Km/Min"✓
    End if

    Return ("")
End Function
```
(3)

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1.2 1.2.1 Private FitnessCalc As clsCalculateSession✓ (1)

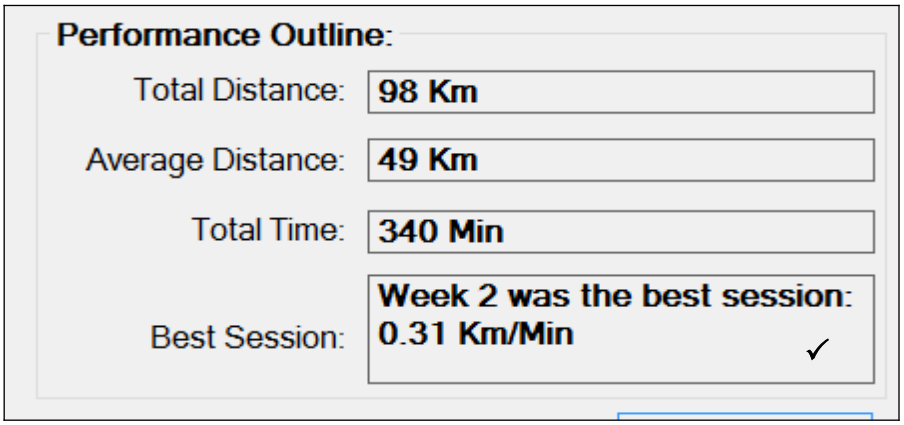
1.2.2 Private Sub bntcalculcate_Click() Handles btnCalculcate.Click

```
If (Not IsNumeric(txtTimeWk1.Text))✓ Or (Not
    IsNumeric(txtDistanceWk1.Text)) _
Or (Not IsNumeric(txtTimeWk2.Text)) Or (Not
    IsNumeric(txtDistanceWk2.Text)) Then
   MessageBox.Show("Time and Distance must be  given as an Integer",
    "Value Type Error", _ MessageBoxButtons.OK, _
      MessageBoxIcon.Error, _
      MessageBoxDefaultButton.Button1)
   txtDistanceWk1.Text = 0
   txtTimeWk1.Text = 0
   txtDistanceWk2.Text = 0
   txtTimeWk2.Text = 0
   txtDistanceWk1.Focus()✓
   Exit Sub
End If
```
(2)

1.2.3
```
FitnessCalc = New✓ clsCalculateSession( _
                        txtTimeWk1.Text,✓ _
      txtTimeWk2.Text✓, _
      txtDistanceWk1.Text✓, _
      _txtDistanceWk2.Text✓)
```
(5)

1.2.4
```
lblBestSession.Text = FitnessCalc.BestSession
lblTotdistance.Text = FitnessCalc.CalDistance() & " Km"
lblAvgDist.Text = FitnessCalc.CalAverageDistance & " Km"✓
lblTotalTime.Text = FitnessCalc.calTotalTime & " Min"✓
```
(2)

1.3



(1)
[25]

QUESTION 2

2.1 2.1.1

```
Private Sub Button1_Click()
      Try
Dim connection As New OleDb.OleDbConnection ✓
Dim sql As New OleDb.OleDbCommand ✓

connection.ConnectionString = ✓"Provider=Microsoft.Jet.OLEDB.4.0;Data
Source=C:\Users\DATABASES\Education.mdb"
connection.Open()✓
```

(4)

2.1.2

```
sql.Connection = connection✓
sql.CommandText = "INSERT INTO
✓learner(learnerId,surname,name,contact,subjectid)
✓VALUES(@studid,@surname,@name,@contact,@subjectid)"✓

Dim stud =✓ New OleDb.OleDbParameter("@studid", txtLearnerID.Text)
Dim surname = New OleDb.OleDbParameter("@surname", txtSurname.Text)
Dim name = New OleDb.OleDbParameter("@name", txtName.Text)
Dim contact =✓New OleDb.OleDbParameter("@contact", txtContact.Text)

Dim subject = New OleDb.OleDbParameter("@subjectid", txtSubjectID.Text)
```

(Any 2 for parameters)

```
 sql.Parameters.Add(studid)✓ ✓
 sql.Parameters.Add(surname)
 sql.Parameters.Add(name)
 sql.Parameters.Add(contact)
sql.Parameters.Add(subjectid)
```

(Any 2 for parameters list)

```
sql.ExecuteNonQuery()✓

        connection.Close()✓
      Catch ex As Exception
        MessageBox.Show(ex.ToString)
      End Try
   End Sub
```

(8)

2.2 2.2.1

```
Private Sub Button1_Click_1() Handles btnSubjectList.Click
 Try
 Dim connection As New OleDb.OleDbConnection
 Dim sql As New OleDb.OleDbCommand✓
```

(1 mark for connect)

```
 Dim table As New Data.DataTable ✓ 1 mark for tabledata
 connection.ConnectionString = "Provider=Microsoft.Jet.OLEDB.4.0;Data
Source= C:\Users\DATABASES\Education.mdb"✓
```

(1 mark for path)

```
connection.Open()✓
```

(4)

2.2.2

```
"SELECT Teacher.TechearID, Teacher.Surname, Learner.Surname,
Subject.SubjectName, Subject.SubjectFee✓ FROM Teacher JOIN Subject ON
Teacher.TechearID = Subject.TechearID✓ JOIN Learner on Learner.SubjectID
= Subject.SubjectID✓ WHERE (((Teacher.TechearID)=@teacher))"✓ (1 mark
for parameter)
Dim subject = New OleDb.OleDbParameter("@teacher", txtTeacherId.Text)✓
sql.Parameters.Add(subject)✓
[OR ANY OTHER VALID SELECT STATEMENT]
```

(6)

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2.2.3

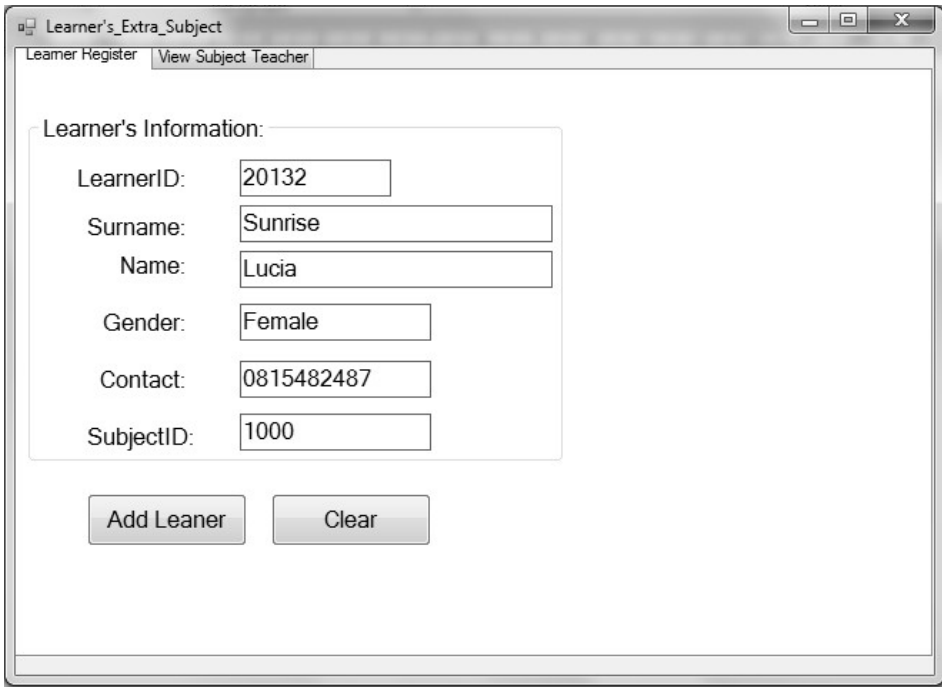
```
            table.Load(sql.ExecuteReader)
        dgvSubject.DataSource = table✓
            connection.Close()

        Catch ex As Exception
            MessageBox.Show(ex.ToString)
        End Try

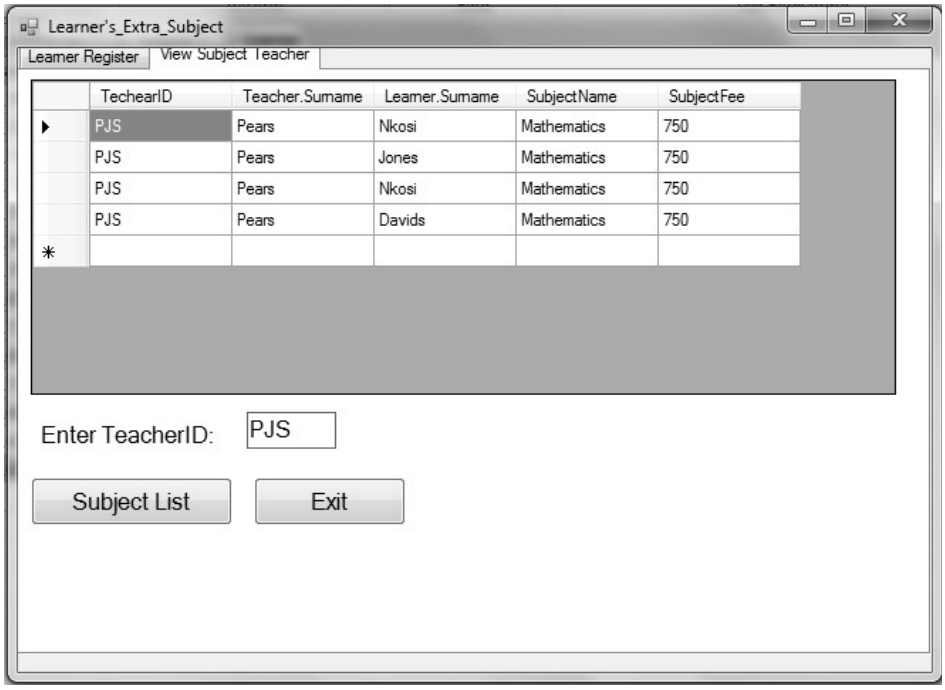
    End Sub
```

(1)

2.3



(1)



(1)
[25]

QUESTION 3

3.1 3.1.1

```
a:link ✓{text-decoration:none;✓
}
```
(2)

3.1.2

```
#nav ✓{text-align:center;✓
        height:30px;
        background-color:#ccc;
        font-weight:bold;✓
}
```
(3)

3.2 3.2.1

```
<html>
<head>
<title<Student Accomodation</title>
<link rel="stylesheet" href="style.css"  type="text/css">✓
</head>
```
(1)

3.2.2

```
<body>
<div id="container">
<div id="header">
  <h1>SK College Course Registration Cost</h1>✓
</div>
```
(1)

3.2.3

```
<div id="nav">
<a href="Home.html">Home</a> | ✓
<a href="Bursary.html">Bursary</a>✓ |
<a href="CollegeSRC.html">SRC</a>✓ |
<a href="Courses.html">Courses</a> | ✓
<a href="Registration.html">Registration</a>✓
</div>
```
(5)

3.2.4

```
<div id="sidebar">✓
    <img src="images/know.jpg" width="150" height="400">✓
</div>
```
(2)

3.2.5

```
<div id="content">
<table border="0" cellspacing="1" cellpadding="2">✓
```
(1)

3.2.6

```
<form name="register">✓
```
(1)

3.2.7

```
<hr>✓ <h2> Registration Process✓ </h2><hr>✓
```
(2)

3.2.8

```
<tr>
<td>Student ID:✓</td><td><input type="text" name="Studid"></td>✓
</tr>
<tr>
<td>Surname:</td><td><input type="text" name="Surname"></td>✓
</tr>
```
(3)

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```
<tr>
<td>Number of  Subjects:✓</td><td><input type="text"
name="noSubjects"></td>
</tr>
```

3.2.9
```
<tr><td>Major Subject:</td><td><Select name="major">✓
<option value="0">Select your Major</option>✓
<option value="1">Software Development</option>
<option value="2">Networking</option>
<option value="3">Analyst</option>
<option value="4">Finance</option> </td>
</tr>
```
(3)

3.2.10
```
<tr>
<td colspan="2"✓><fieldset ✓><legend>Select
Residence</legend>
<input type="radio" name="campus">ON-Campus<br>✓
<input type="radio" name="campus">OFF-Campus
</fieldset>
</td>
</tr>
```
(3)

3.2.11
```
<tr>
<td><input type="button" value="Calculate Costs"
onclick✓="computer()"✓></td>
```
(2)

3.2.12
```
<td><input type="reset"  ✓value="Clear Text"> </td></tr>
```
(1)

3.2.13
```
<tr><td> <p id="regCost"></td></tr> ✓
</table>
</div>
```
(1)

3.2.14 Printout of Webpage (1)

[32]

QUESTION 4

4.1 4.1.1
```
<script type="text/javascript">
function compute()✓
{
```
(1)

4.1.2
```
var nrSubject=0; ✓
var totalCost = 0; ✓
var campus;  ✓
```
(3)

4.1.3
```
var campus = document.getElementById("campus").checked✓;
nrSubject = document.getElementById("noSubjects").value;✓
```
(2)

4.1.4
```
If (Number✓ (nrSubject) ✓>0) ✓
{
```
(3)

4.2 4.2.1
```
        if(campus == true✓)
            {
              totalCost✓ = (nrSubject * 750) ✓ + 8500,00;✓
            }
```
(4)

4.2.2
```
        Else {
                totalCost ✓= (nrSubject * 750)✓
              }
```
(2)

4.2.3
```
        document.getElementById("regCost").value ✓= "Total course
        cost is R" + totalCost.toFixed;✓
} //end  if
} //end of function
</script>
```
(2)

4.2.4 Printout of webpage showing results of function ✓ (1)

[18]

TOTAL: 100

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